



NeuroSpace

Complete evaluation export

Job ID	layout-test
Status	completed
Result timestamp	2026-05-13T21:32:37.402087
PDF generated	2026-05-15 21:13:53

Pipeline Configuration

Parameter	Value
model_types	rnn, qrnn
comparison_mode	mi
feature_selector	mi
manual_overrides.rnn_epochs	30
manual_overrides.rnn_batch_size	64
manual_overrides.qrnn_epochs	15
manual_overrides.qrnn_batch_size	16
num_classes	6
class_names	1, 2, 3, 4, 5, 6
qrnn_backend	qiskit
noise_enabled	False
noise_level	0.0000
mitigation_enabled	False
mitigation_runs	1

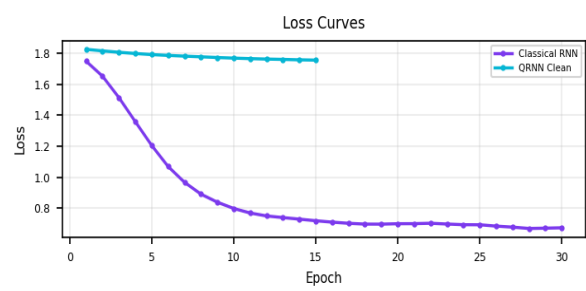
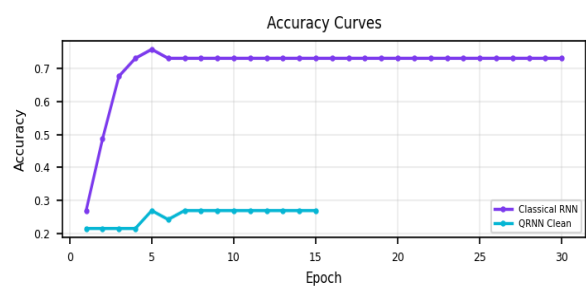
Best Model

Winner	RNN Accuracy	QRNN Accuracy	Improvement
RNN	81.08%	28.38%	52.70%

Evaluation Summary

Model	Accuracy	Precision	Recall	F1 Score	Loss
Classical RNN	81.08%	81.14%	81.08%	80.05%	0.6721
QRNN Clean	28.38%	22.31%	28.38%	22.17%	1.7577

Learning Curves



Classical RNN

Metric	Value
Accuracy	81.08%
Precision	81.14%
Recall	81.08%
F1 Score	80.05%
Loss	0.6721
Train Loss	0.5927
Val Loss	0.6721
Train Accuracy	81.18%
Val Accuracy	72.97%
Train Size	255
Val Size	37
Test Size	74

Classical RNN Confusion Matrix

Actual / Predicted	Class 1	Class 2	Class 3	Class 4	Class 5	Class 6
Class 1	23	0	0	0	0	0
Class 2	0	3	0	5	1	3
Class 3	0	0	15	0	0	0
Class 4	0	2	0	6	0	2
Class 5	0	0	0	0	10	0
Class 6	0	1	0	0	0	3

Classical RNN Classification Report

Class	Precision	Recall	F1 Score	Support
1	100.00%	100.00%	100.00%	23.0000
2	50.00%	25.00%	33.33%	12.0000
3	100.00%	100.00%	100.00%	15.0000
4	54.55%	60.00%	57.14%	10.0000
5	90.91%	100.00%	95.24%	10.0000
6	37.50%	75.00%	50.00%	4.0000
macro avg	72.16%	76.67%	72.62%	74.0000
weighted avg	81.14%	81.08%	80.05%	74.0000
Accuracy			81.08%	

Classical RNN Epoch History

Epoch	Train Loss	Val Loss	Train Acc	Val Acc
1	1.9031	1.7504	15.69%	27.03%
2	1.5239	1.6535	39.22%	48.65%
3	1.2855	1.5149	60.78%	67.57%
4	1.1040	1.3589	73.73%	72.97%
5	0.9299	1.2049	76.47%	75.68%
6	0.8834	1.0687	76.08%	72.97%
7	0.8142	0.9665	79.61%	72.97%
8	0.7798	0.8900	76.86%	72.97%
9	0.7488	0.8378	78.43%	72.97%
10	0.7321	0.7971	77.65%	72.97%
11	0.7132	0.7679	79.22%	72.97%
12	0.7210	0.7489	78.82%	72.97%
13	0.6922	0.7382	76.47%	72.97%
14	0.6793	0.7281	78.43%	72.97%
15	0.6708	0.7177	80.00%	72.97%
16	0.6461	0.7087	81.96%	72.97%
17	0.6462	0.7013	79.61%	72.97%
18	0.6534	0.6960	80.39%	72.97%
19	0.6190	0.6957	80.78%	72.97%
20	0.6563	0.6985	78.43%	72.97%
21	0.6063	0.6989	79.61%	72.97%
22	0.6128	0.7015	80.00%	72.97%
23	0.5909	0.6965	80.78%	72.97%
24	0.6331	0.6921	77.25%	72.97%
25	0.6164	0.6917	78.82%	72.97%
26	0.6027	0.6831	81.57%	72.97%
27	0.5887	0.6764	79.22%	72.97%
28	0.5784	0.6677	81.96%	72.97%
29	0.5891	0.6697	80.39%	72.97%
30	0.5927	0.6721	81.18%	72.97%

QRNN Clean

Metric	Value
Accuracy	28.38%
Precision	22.31%
Recall	28.38%
F1 Score	22.17%
Loss	1.7577
Train Loss	1.7065
Val Loss	1.7577
Train Accuracy	35.69%
Val Accuracy	27.03%
Train Size	255
Val Size	37
Test Size	74

QRNN Clean Confusion Matrix

Actual / Predicted	Class 1	Class 2	Class 3	Class 4	Class 5	Class 6
Class 1	4	0	17	2	0	0
Class 2	0	8	3	0	0	1
Class 3	0	6	9	0	0	0
Class 4	0	8	1	0	1	0
Class 5	7	0	2	0	0	1
Class 6	0	1	2	0	1	0

QRNN Clean Classification Report

Class	Precision	Recall	F1 Score	Support
1	36.36%	17.39%	23.53%	23.0000
2	34.78%	66.67%	45.71%	12.0000
3	26.47%	60.00%	36.73%	15.0000
4	0.00%	0.00%	0.00%	10.0000
5	0.00%	0.00%	0.00%	10.0000
6	0.00%	0.00%	0.00%	4.0000
macro avg	16.27%	24.01%	17.66%	74.0000
weighted avg	22.31%	28.38%	22.17%	74.0000
Accuracy			28.38%	

QRNN Clean Epoch History

Epoch	Train Loss	Val Loss	Train Acc	Val Acc
1	1.8160	1.8274	22.35%	21.62%
2	1.8011	1.8174	21.96%	21.62%
3	1.7879	1.8083	21.96%	21.62%
4	1.7786	1.8005	21.96%	21.62%
5	1.7667	1.7935	28.24%	27.03%
6	1.7574	1.7882	29.41%	24.32%
7	1.7521	1.7834	31.37%	27.03%
8	1.7456	1.7791	32.55%	27.03%
9	1.7368	1.7745	33.73%	27.03%
10	1.7306	1.7707	36.08%	27.03%
11	1.7254	1.7674	36.08%	27.03%
12	1.7226	1.7644	36.86%	27.03%
13	1.7141	1.7623	37.65%	27.03%
14	1.7105	1.7595	36.08%	27.03%
15	1.7065	1.7577	35.69%	27.03%